FIFTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING AND TECHNOLOGY

INDUSTRIAL AUTOMATION

MODEL QUESTION PAPER – SET-1

Time: 3 hours Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

		Module Outcome	Cognitive level
1	The angle at which SCR starts conducting is called ……..	M1.03	U
2	Draw the symbol of IGBT and name the terminals.	M1.01	R
3	A device that converts fixed DC I/p to variable DC o/p voltage is called ………..	M2.04	U
4	………is a device that provides backup power when regular power source fails or voltage drops to an unacceptable level.	M3.04	U
5	List any 2 disadvantages of Stator Voltage Control method.	M3.02	R
6	PLC means …………………………………………..	M4.01	R
7	In a ladder diagram normally open contacts in parallel represents …… function.	M4.03	U
8	Write any two applications of Induction heating.	M3.03	R
9	What is the reason for controlled rectifiers.	M2.01	R

PART B

II. Answer any Eight questions from the following

(8 x 3 = 24 Marks)

		Module Outcome	Cognitive level
1	Develop ladder logic for Boolean expression Y=(AB+C)D	M4.04	A
2	Define the term Holding current and Latching current	M1.01	R
3	Draw the circuit of Half wave controlled rectifier with resistive load and its output response.	M2.01	U
4	List any 3 advantages of PLC over relays.	M4.02	R
5	Develop ladder diagram for a two way switch stair case lamp.	M4.04	A
6	List any 3 comparisons between AC and DC drives	M3.01	R
7	State the principle of Dielectric heating.	M3.03	R

8	Draw the circuit and output response of a step down chopper.	M2.04	U
9	With a neat sketch explain the two-transistor analogy of SCR	M1.02	U
10	Which UPS is better, ON-line or OFF-line and why.	M3.04	R

PART C

III. Answer all questions from the following **(6 x 7 = 42 Marks)**

		Module Outcome	Cognitive level
1 a.	With a neat sketch describe the structure of SCR	M1.01	U
1 b.	List any 6 applications of SCR	M1.01	R
	OR		
2	Draw the circuit and explain the working of UJT triggering of SCR	M1.03	U
3	With a neat sketch explain the working of power MOSFET.	M1.01	U
	OR		
4	With circuit diagram explain the working of class-C commutation technique.	M1.04	U
5	With circuit diagram explain the working of Jone's chopper.	M2.04	U
	OR		
6 a.	Explain the working of a Series inverter with circuit diagram	M2.02	U
6 b.	List any 2 differences between a series inverter and a parallel inverter.	M2.02	R
7	Explain with necessary diagrams about the operation of a Centre Tapped Full Wave controlled rectifier with RL load.	M2.01	U
	OR		
8	With a neat circuit diagram explain the working of a step-up Cyclo Converter.	M2.03	U
9	Draw the block diagram and explain the working of an ON-line UPS	M3.04	U
	OR		
10	List and explain any 3 types of resistance welding.	M3.03	U
11	Explain about Program Control instructions used in PLC	M4.03	U
	OR		
12	List the main parts of a PLC and write the function of each part.	M4.01	U